AgriTech & Environment Sustainability

Project Name: Crop Ripeness Analyzer





The Problem: Riya's uncle is a farmer who grows many fruits and vegetables. 🍅 He finds it difficult to know which crops are ready to harvest because some look very similar. Checking every crop by hand takes a lot of time and can lead to mistakes. Riya wants to help him by making a smart system that can check the crop's colour.

🎯 **What You Will Design:** Students will build a smart crop ripeness analyzer using a color sensor and Arduino Uno. The sensor will detect the color of a fruit or crop and display the result on an OLED screen, while an LED provides a visual indication.



🎨 What is a Color Sensor?

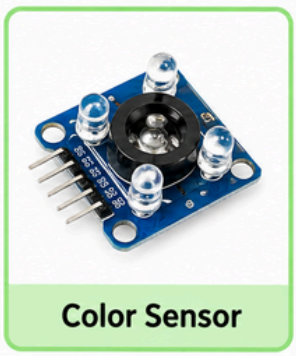
A color sensor is a device that identifies different colors by detecting light reflected from an object. It helps machines recognize the color of fruits, vegetables, and many other objects.

Example: 🍎 Just like you can tell whether an apple is red or green using your eyes, a color sensor can also identify the color of the apple.



Components Required







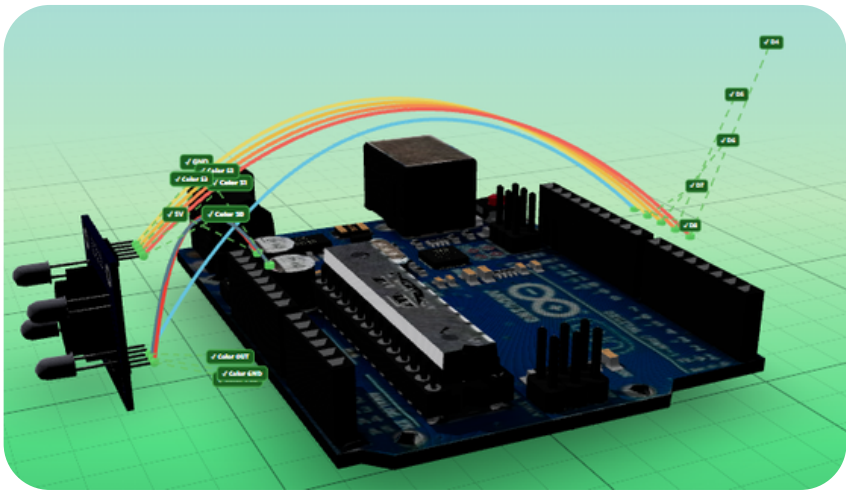




Circuit Designing

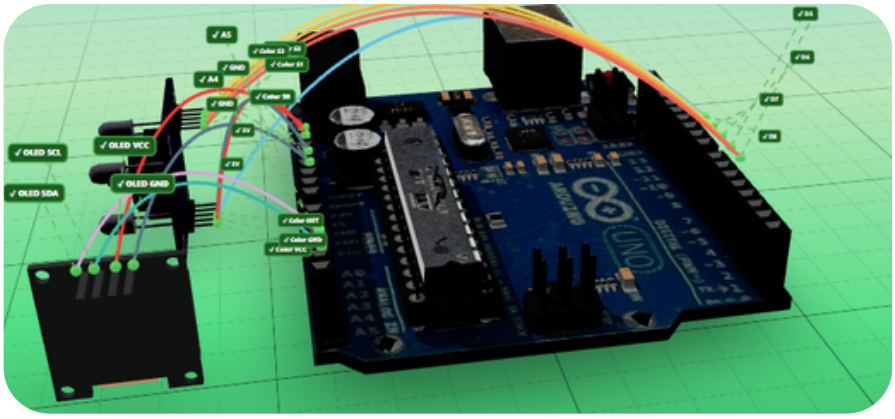
Step 1: Connect the color sensor to the Arduino Uno.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
S0 Pin	Digital Pin 4
S1 Pin	Digital Pin 5
S2 Pin	Digital Pin 6
S3 Pin	Digital Pin 7
OUT Pin	Digital Pin 8



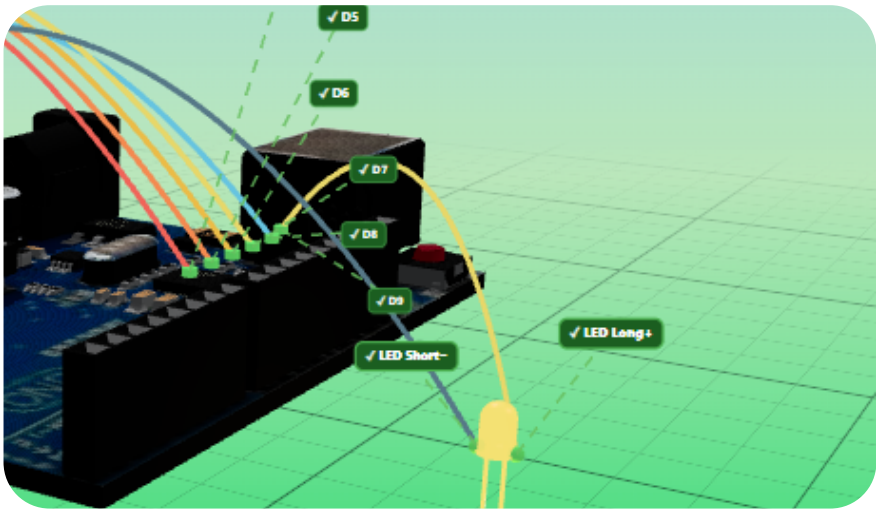
2. Connect the OLED display to the Arduino Uno.

OLED Pins	Arduino Pins
Vcc Pin	5v Pin
GND Pin	GND Pin
SDA Pin	Analog A4 Pin
SCL Pin	Analog A5 Pin



3. Connect the LED to the Arduino Uno.

LED Pins	Arduino Pins
Long Leg	Digital Pin 9
Short Leg	GND Pin



Project Coding

1. Make the code to initialize the pins.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// Pin definitions
#define S0 4
#define S1 5
#define S2 6
#define S3 7
#define sensorOut 8
#define ledPin 9
```

2. Make code to do the OLED display settings and define the crop ripeness color values.

```
// OLED display settings
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// Predefined color values for ripeness
const int ripeRed = 100; // Example value
const int ripeGreen = 50; // Example value
const int ripeBlue = 50; // Example value
```

3. Make code to initialize the serial communication and the OLED display.

```
31  void setup() {
32    // Initialize serial communication
33    Serial.begin(9600);
34
35    // Initialize OLED display
36    if (!display.begin(SSD1306_I2C_ADDRESS, 0x3C)) {
37      Serial.println(F("SSD1306 allocation failed"));
38      for (;;);
39    }
40    display.clearDisplay();
41    display.setTextSize(1);
42    display.setTextColor(SSD1306_WHITE);
```

4. Make code to intialize the input and the output pins.

```
44    // Initialize pins
45    pinMode(S0, OUTPUT);
46    pinMode(S1, OUTPUT);
47    pinMode(S2, OUTPUT);
48    pinMode(S3, OUTPUT);
49    pinMode(sensorOut, INPUT);
50    pinMode(ledPin, OUTPUT);
51
52    // Set frequency scaling to 20%
53    digitalWrite(S0, HIGH);
54    digitalWrite(S1, LOW);
55  }
```

5. Make code to read the color values and print them in the format of RGB color.

```
57  void loop() {
58    int red = readColor(S2, S3, LOW, LOW);
59    int green = readColor(S2, S3, HIGH, HIGH);
60    int blue = readColor(S2, S3, LOW, HIGH);
61
62    Serial.print("R: "); Serial.print(red);
63    Serial.print(" G: "); Serial.print(green);
64    Serial.print(" B: "); Serial.println(blue);
65
66    display.clearDisplay();
67    display.setCursor(0, 0);
```

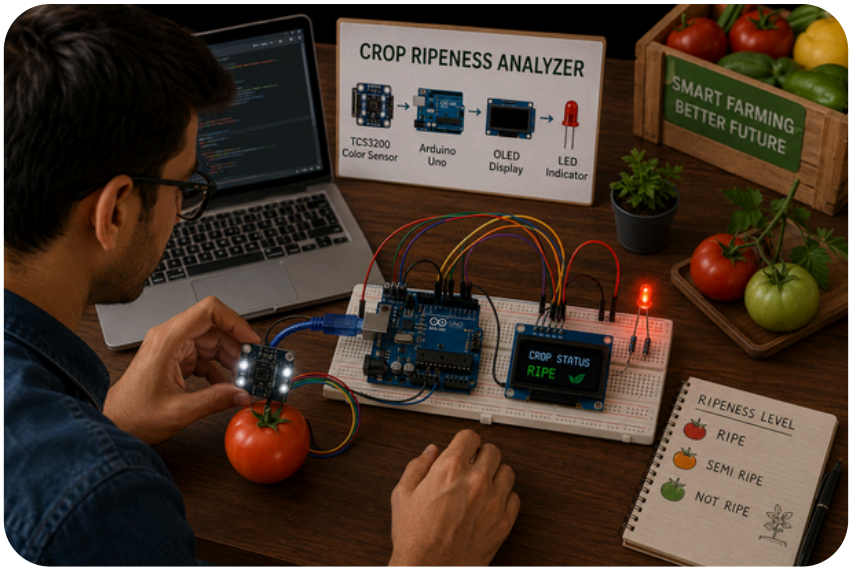
6. Make code to set the conditions to check if the crop is ripped or not.

```
69      if (isRipe(red, green, blue)) {
70        display.println("Ripe Crop");
71        digitalWrite(ledPin, HIGH);
72      } else {
73        display.println("Not Ripe Yet");
74        digitalWrite(ledPin, LOW);
75      }
76
77      display.display();
78      delay(1000);
79    }
```

7. Make code to print the final color values.

```
81  int readColor(int s2, int s3, int s2State, int s3State) {
82    digitalWrite(s2, s2State);
83    digitalWrite(s3, s3State);
84    return pulseIn(sensorOut, LOW);
85  }
86
87  bool isRipe(int red, int green, int blue) {
88    return (red >= ripeRed && green <= ripeGreen && blue <= ripeBlue);
89  }
```

Project Output





🌾 Where Is This Used in Real Life?

Farmers can use this project to check whether fruits and crops are ready to harvest by looking at their color. It can be used in farms, fruit gardens, and food sorting centers to save time and reduce mistakes.

Learning Outcomes

- The students will be able to understand how a color sensor detects different colors.
- The students will be able to write and upload code to the Arduino Uno.
- The students will be able to explain how technology helps farmers improve crop monitoring.



Section 1: Multiple Choice Questions

1. What problem does Riya want to solve?

a) Watering all the crops
b) Checking which crops are ready to harvest
c) Planting new trees
d) Counting the number of farmers

2. Which part helps the project detect the color of the crop?

a) Color Sensor
b) Jumper Wire
c) OLED Display
d) Arduino USB Cable

3. Riya wants the farmer to easily see the crop status. Which part should she use?

a) Jumper Wire
b) Color Sensor
c) OLED Display
d) GND Pin

Section 2: True and False Statements

1. Checking every crop by hand can take a lot of time.....

2. The color sensor helps the project detect the color of a crop.....

3. The Crop Ripeness Analyzer is made to grow crops automatically.....